

Sharon Zhang

CONTACT

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INTERESTS

Agentic content creation, verification for visual content, programmatic visual generation.

EDUCATION

2021 - Current	Stanford University
	PhD in Computer Science Advisor: Maneesh Agrawala
2017 - 2021	Princeton University
	A.B. in Mathematics with honors · Minor in Computer Science Thesis: <i>Contextual Bias and Interpretability in Visual Classification</i> Advisor: Olga Russakovsky

RESEARCH EXPERIENCE

2020 - 2021	Visual AI Lab, <i>Princeton University</i>
	Advisor: Olga Russakovsky
2020	Department of Mathematics, <i>Princeton University</i>
	Advisor: Amit Singer
2019	SMALL REU, <i>Williams College</i>
	Advisor: Susan Loepp Undergraduate researcher in the 2019 Commutative Algebra cohort.

INDUSTRY EXPERIENCE

2025	Research Intern, <i>Roblox</i>
	Contributed to research and development on the Foundation AI team for LLM-assisted functional object and scene generation. Developed verification pipelines for evaluating generation results and populating knowledge bases for in-context examples.
2024	Research Intern, <i>Roblox</i>
	Designed and implemented a procedural generation pipeline for creating obstacle course games as part of a larger project on training RL agents for helping assess obstacle course game difficulty.
2021	Research Intern, <i>Google</i>
	Worked on layered video representations with the Visual Dynamics team. Hosted by Jonathan Huang and Vivek Rathod.
2020	Software Engineering Intern, <i>Google</i>
	Hosted by the Android Camera Machine Intelligence team. Designed and implemented an open-sourced API for Android developers to incorporate Google motion photos.

PREPRINTS

Sharon Zhang, R. Kenny Jones, Jiajun Wu, Maneesh Agrawala. "Aggregating LLM-Based Weak Verifiers for Spatial Layout Generation." 2026. <https://arxiv.org/abs/2606.05268>.

PUBLICATIONS

Sharon Zhang, Jiaju Ma, Jiajun Wu, Daniel Ritchie, Maneesh Agrawala. "Editing Motion Graphics Video via Motion Vectorization and Transformation." *SIGGRAPH Asia*, 2023.

Chen Geng*, Hong-Xing Yu*, **Sharon Zhang**, Maneesh Agrawala, Jiajun Wu. "Tree-Structured Shading Decomposition." *International Conference on Computer Vision (ICCV)*, 2023.

Sunnie S. Y. Kim, **Sharon Zhang**, Nicole Meister, Olga Russakovsky. "[Re] Don't Judge an Object By It's Context: Overcoming Contextual Bias." Machine Learning Reproducibility Challenge 2020. In *ReScience C Journal*.

Sharon Zhang, Amit Moscovich, Amit Singer. "Product Manifold Learning." *Artificial Intelligence and Statistics (AISTATS)*, 2021.

Erica Barrett*, Emil Graf*, Kimball Strong*, **Sharon Zhang***, S. Loepp. "Cardinalities of Prime Spectra of Precompletions." *AMS Contemporary Mathematics: "Commutative Algebra: 150 Years with Roger and Sylvia Wiegand."* **773** (2021), 133–152.

Erica Barrett*, Emil Graf*, Kimball Strong*, **Sharon Zhang***, S. Loepp. "Structure of spectra of precompletions." *Rocky Mountain J. Math.* **50** (2020), no. 6, 1965–1988.

ONGOING WORK

My current work covers verification for 2D/3D layout generation and RL-based tools for 3D game design. Feel free to reach out for more details.

AWARDS

2023 - 2026	NSF Graduate Research Fellowship
2023	Ford Foundation Fellowship Honorable Mention
2023	Paul & Daisy Soros Fellowship Finalist
2022, 2023, 2025	Brown Institute for Media Innovation Magic Grant
2021	Middleton Miller '29 Prize, <i>awarded for best independent work in mathematics</i>
2021	NSF Graduate Research Fellowship Honorable Mention

ACTIVITIES & SERVICE

	Reviewer for AISTATS, Eurographics, CHI, DIS
2023	Organizer, Stanford Graphics Café
2020	Teaching Assistant, Princeton University COS 429: Computer Vision
2019	Peer Tutor, Princeton University MAT 215: Honors Analysis
2019	Peer Tutor, Princeton University COS 126: Introduction to Computer Science

COMPUTER SKILLS

Languages	Python, Java, HTML/CSS, C/C++, Lua, bash shell scripting
Software & Tools	JAX, PyTorch, TensorFlow, Git, CVX, L ^A T _E X, Google Cloud, Adobe Creative Suite

June 5, 2026